

STATE-DEPENDENT SPECIATION AND EXTINCTION (SSE) MODELS

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PhylogeogRaphy Workshop

Day 2

1 WHAT WE HAVE DONE SO FAR

2 LIKELIHOOD FOR BiSSE

3 ANCESTRAL STATE INFERENCE FOR SSE MODELS

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KOLMOGOROV EQUATIONS

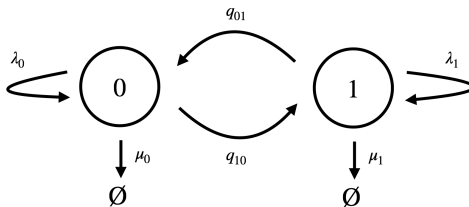
The Yule and Birth-death processes treat different lineages independently, and also assume that all lineages are identical in terms of their birth/death rates.

Next, we are going to consider branching processes that allow for lineages to differ in terms of their birth/death rates.

Quantifying the probability of a tree in this case is more complicated, but the backward equation approach comes to the rescue.

EXTENSION TO MULTIPLE STATES

The **binary state-dependent speciation and extinction (BiSSE)** model extends the birth-death model. Lineages reproduce and go extinct, but possibly at different rates depending on their state. They can migrate/mutate between two states:



AN EXAMPLE BiSSE TREE:

The states in BiSSE are just like the Markov chain models for traits we saw in ace and simmap, except that now the states are correlated with branching events in the tree:

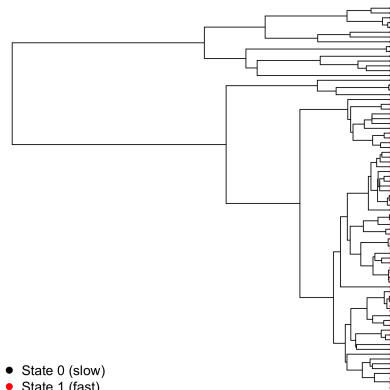


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BISSSE EQUATIONS (ON A SINGLE BRANCH)

A couple of points to recall: there are at least 3 distinct notions of “backward” that are prevalent in the phylodynamics literature, which can be confusing:

1. Conventionally, time increments backward into the past¹
2. We derive backward differential equations by expanding near initial states/conditioning on the past (“jumps near the initial condition”).
3. Phylodynamic likelihoods are obtained using “Felsenstein’s Pruning Algorithm”, which constructs tree likelihoods recursively from the tips to the root, i.e., “backward”.

Phylodynamic likelihoods are “backwards” in each of these ways.

¹Wayne P Maddison, Peter E Midford, and Sarah P Otto. “Estimating a binary character’s effect on speciation and extinction”. In: *Systematic biology* 56.5 (2007), pp. 701–710.

BISSE EQUATIONS

The state space of possible trees is much more complicated now.

Before, we could treat sister lineages as independent and identically distributed, in terms of their birth/death events

This isn't true anymore: with different state changes possible in these new models, sister lineages will remain independent of each other, but we can no longer treat them identically.

Our backward equation formalism from the birth-death-sampling-through-time model will help us tremendously.

CONNECTING THESE PROCESSES TO TREE TOPOLOGIES

Births in these stochastic processes correspond to bifurcations in trees.

Deaths/extinctions mean that some lineages disappear/terminate before our stopping time

Migrations in the stochastic process will correspond to changes of state on branches of a phylogeny

Lineages in different states experience branching/extinction at different rates

BISSE EQUATIONS

In this state space of trees with time-dependent states along lineages, there is no tractable analog of the Chapman-Kolmogorov equation for us to work with (as far as I am aware):

$$p_{i,j}(t+s) = \sum_k p_{i,k}(s)p_{k,j}(t).$$

Instead, we need to define backward variables similar to the sampling-through-time birth-death model².

However: let's assume perfect sampling at the present ($t = 0$) to start. We can always make our lives harder later.

²Tanja Stadler. "Sampling-through-time in birth-death trees". In: *Journal of theoretical biology* 267.3 (2010), pp. 396–404.

STUFF THAT CAN HAPPEN IN BiSSE

A lineage in state i can

- split into two lineages at rate λ_i ,
- die at rate μ_i ,
- switch to state $j \neq i$ at rate q_{ij} .

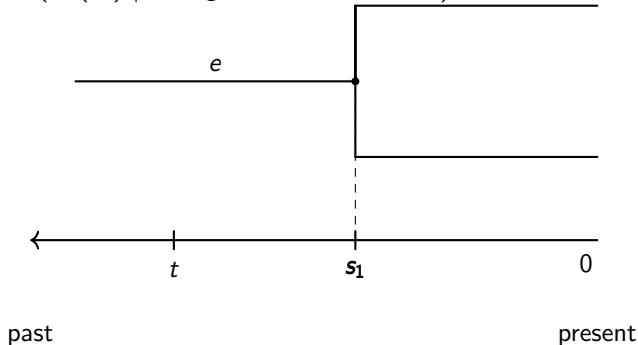
Not so terrible... but the challenge is that an infinite number of events can occur before the process ends at (time $t = 0$).

PROBABILITY OF A SUBTREE

Recall the sampling-through-time birth-death model:

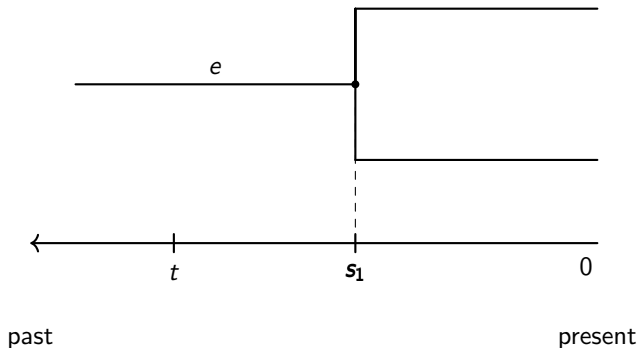
$\mathcal{T}_e(s_1)$: the sub-tree descending from edge e at time s_1

$g_e(t) = \mathcal{P}(\mathcal{T}_e(s_1) \mid \text{lineage } e \text{ alive at time } t)$



We make a very minor modification: we recognize that edge e can reside in state i .

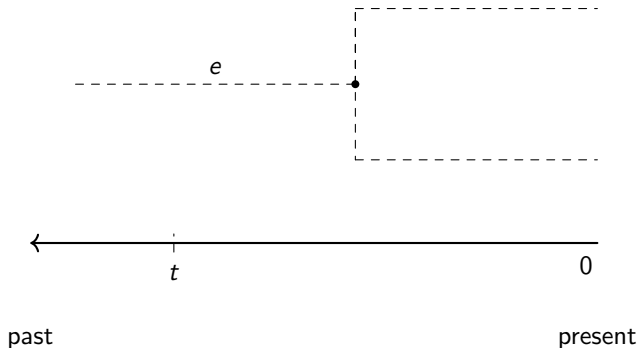
PROBABILITY OF A SUBTREE



We consider the quantities $\mathcal{T}_e(s_1)$: the sub-tree descending from edge e at time s_1 (same as before)

$$g_{e_i}(t) = \mathcal{P}(\mathcal{T}_e(s_1) \mid \text{lineage } e \text{ alive and in state } i \text{ at time } t)$$

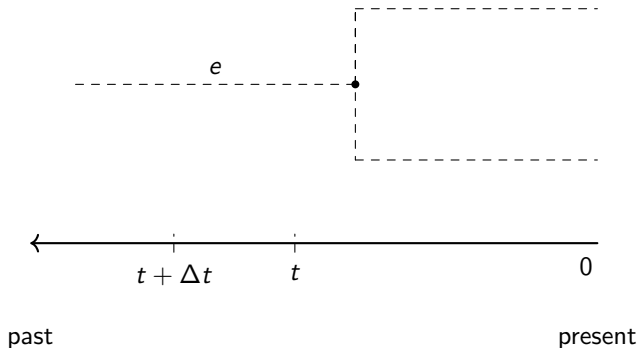
PROBABILITY A LINEAGE GOES EXTINCT



We also have to consider state-dependent extinction: let

$$E_{e_i}(t) = \mathcal{P}(\text{lineage } e \text{ leaves no descendants} \mid \text{lineage } e \text{ alive and in state } i \text{ at time } t)$$

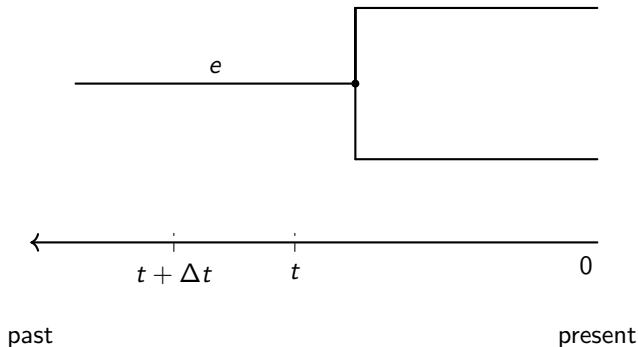
PROBABILITY A LINEAGE GOES EXTINCT



Derive the backward equation:

$$E_{e_i}(t + \Delta t) = \mu_i \Delta t \times 1 + \sum_j q_{ij} \Delta t E_{e_j}(t) + 2\lambda_i \Delta t E_{e_i}(t)^2 + \left(1 - (\lambda_i + \mu_i + \sum_j q_{ij}) \Delta t\right) E_{e_i}(t)$$

DRIVE THE BACKWARD EQUATION FOR g_{e_i}



Derive the backward equation for g_{e_i} :

$$g_{e_i}(t + \Delta t) = \left(\sum_j q_{ij} \Delta t \right) g_{e_j}(t) + 2(\lambda_i \Delta t) g_{e_i}(t) E_{e_i}(t) + \left(1 - (\lambda_i + \mu_i + \sum_j q_{ij}) \Delta t \right) g_{e_i}(t).$$

BiSSE (ACTUALLY, MuSSE) EQUATIONS

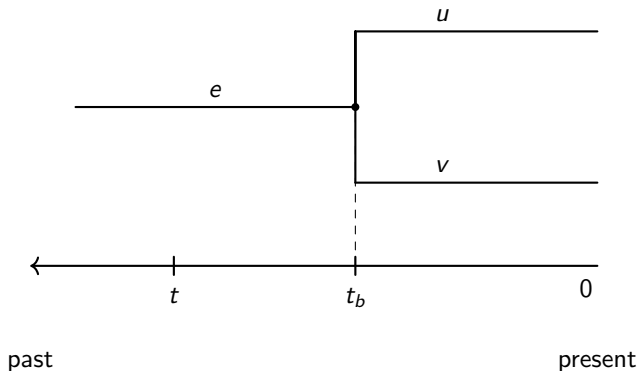
We then apply the standard trick of rearranging terms, dividing by Δt , and taking the limit $\Delta t \rightarrow 0$:

$$\begin{aligned}\frac{dg_{e_i}}{dt} &= - \left(\lambda_i + \mu_i + \sum_j q_{ij} \right) g_{e_i}(t) + \sum_j q_{ij} g_{e_j}(t) \\ &\quad + 2\lambda_i g_{e_i} E_{e_i}, \\ \frac{dE_{e_i}}{dt} &= - \left(\lambda_i + \mu_i + \sum_j q_{ij} \right) E_{e_i}(t) + \sum_j q_{ij} E_{e_j} \\ &\quad + 2\lambda_i E_{e_i}^2.\end{aligned}$$

Initial conditions for edges are determined in the same way as the sampling-through-time birth-death model.

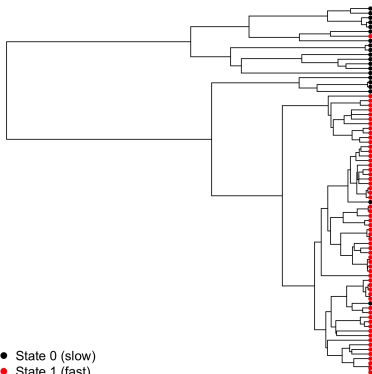
INITIAL CONDITIONS AT BRANCH POINTS:

Suppose we have integrated solutions up two sister lineages u and v , and have $g_{u_i}(t_b^-)$ and $g_{v_i}(t_b^-)$ calculated for all states i . How do we proceed at the bifurcation point at time t_b ?



Answer: we set $g_{e_i}(t_b^+) = \lambda_i g_{u_i}(t_b^-) g_{v_i}(t_b^-)$ as our initial condition for g_{e_i} , and integrate up edge e . Easy!

POSTORDER TRAVERSAL



The $g_{e_i}(0)$ are initialized at the tips, and equal either 0 or 1 because we assume we observe tip states perfectly.

Then the algorithm rips along from the tips to the root (“postorder traversal”)

TREE PROBABILITY/LIKELIHOOD

Eventually we end up at the root node R of the entire tree $\mathcal{T}$. The root R is at time T (confused yet?). We are left with terms

$$g_{R_i}(T) = g_{u_i}(T)g_{v_i}(T)\lambda_i,$$

for each state i .

In a slight abuse of notation (because R is a node rather than an edge), $g_{R_i}(T) = \mathcal{P}(\mathcal{T} \mid \text{root in state } i \text{ at time } T)$.

At this point, we have finally come full circle to **ancestral state inference**. We could select a point estimate for the most likely root state as

$$\hat{y}_R = \operatorname{argmax}_i \{g_{R_i}(T)\pi_i\},$$

where π_i is a prior distribution on states that could be informed by the model parameters and structure.

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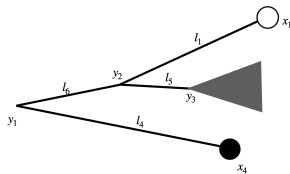
BASIC OVERVIEW OF THE SSE PROBABILITY CALCULATION:

The overall calculation strategy for the SSE tree probabilities goes like this:

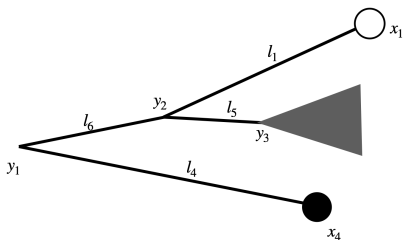
- Start at the tips and integrate up edges to get probabilities of subtrees conditional on edge states
- average these conditional probabilities when they meet at a node
- upon reaching the root node, average the conditional probabilities to obtain the probability of the tree

PRUNING

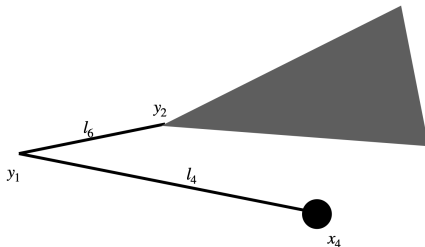
We start at the tips:



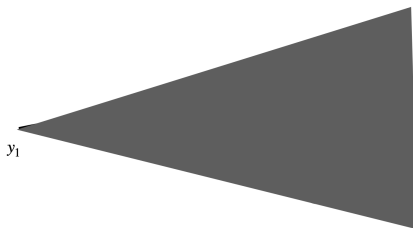
PRUNING



PRUNING



PRUNING



And work our way up. Seem familiar?

The calculations for likelihoods of simple Markov models we saw earlier in the workshop function in exactly this way

THE PRUNING ALGORITHM (DYNAMIC PROGRAMMING)

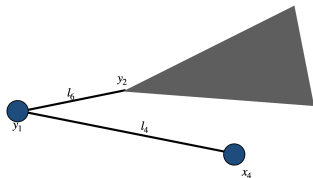
The dynamic programming (pruning) algorithm gives us a likelihood for the tree **conditional** on the root state

We could either select the most likely root state using argmax , or we could sample a root state using the Multinomial/Categorical distribution

For the simple models at the beginning of this workshop (in ace/simmap), we saw that there was a natural way to implement a joint reconstruction with a pre-order traversal (from root to tips)

JOINT RECONSTRUCTION IN ACE/SIMMAP

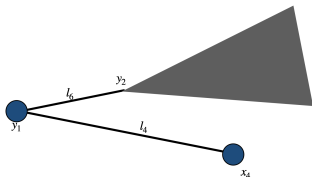
In ace/simmap, we moved from root to tips:



Assume we have already chosen ancestral states for the root y_1 using either argmax (for Maximum Likelihood), or by sampling (e.g., stochastic character mapping). We now want to choose the ancestral state for node y_2 .

The pruning/dynamic programming/postorder traversal algorithm gave us $L_{y_2}(z)$, the probabilities of the sub-tree descending from y_2 , conditional on y_2 being in state z

JOINT RECONSTRUCTION



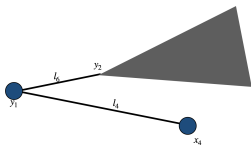
However, $L_{y_2}(z)$ only uses information about the sub-tree from y_2 , so $\operatorname{argmax}_z L_{y_2}(z)$ would **not** give us a joint reconstruction estimate for node y_2

We resolve this by using 2 pieces of information: (1) we have an estimate for the root state $\hat{y}_3$;

(2) along the edge l_6 , we know how to calculate transition probabilities, $p_{xy}(l_6)$

JOINT RECONSTRUCTION

We evaluate $p_{\hat{y}_1, z}(l_6)L_{y_2}(z)$ for each z , and select z using either argmax or sampling



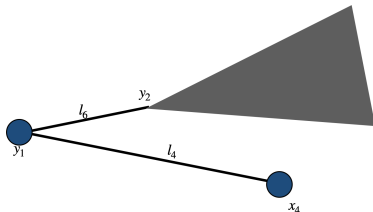
The two quantities being multiplied above are fundamentally different objects:

- $L_{y_2}(z) = \mathcal{P}(\mathcal{T}_{y_2} \mid y_2 = z)$
- $p_{\hat{y}_1, z}(l_6) = \mathcal{P}(y_2 = z \mid y_1 = \hat{y}_1, l_6)$

In one expression, z is the conditioning variable, and in the other expression z is the conditioned variable.

As we've seen, quantities with conditioning variables ($F(\cdot \mid X)$) are naturally described by **backward equations**. **Forward equations** work for quantities with conditioned variables ($F(X \mid \cdot)$)

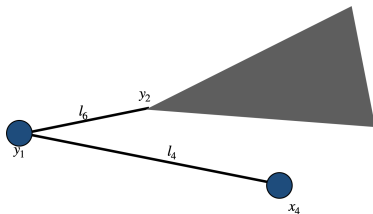
JOINT RECONSTRUCTION



In BiSSE-type models, we have seen that calculating an object like $L_{y_2}(z) = \mathcal{P}(\mathcal{T}_{y_2} \mid y_2 = z)$ requires integrating **backward equations** along edges from tips to root

We need to find an analogous quantity to $p_{\hat{y}_1, z}(l_6) = \mathcal{P}(y_2 = z \mid y_1 = \hat{y}_1, l_6)$ for SSE models.

DERIVING A FORWARD EQUATIONS... FOR WHAT?

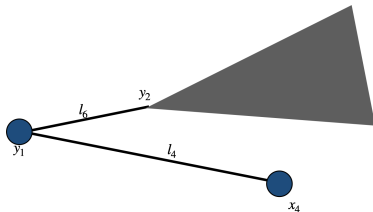


Let t_{y_2} be the time corresponding to node y_2 .

In our SSE models, we have worked out that

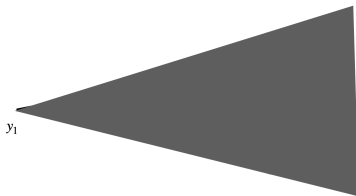
$L_{y_2}(z) = g_{e_z}(t_{y_2}) = \mathcal{P}(\mathcal{T}_{y_2} \mid y_2 = z, t_{y_2})$. The $g_{e_z}(t_{y_2})$ gets built up by first integrating up edges from tips and combining terms using the pruning algorithm

DERIVING A FORWARD EQUATIONS... FOR WHAT?



The quantities $g_{e_z}(t_{y_2}) = \mathcal{P}(\mathcal{T}_{y_2} \mid y_2 = z, t_{y_2})$ do not give us information about the transition probability from the root state to the node y_2 . They just tell us the conditional probability of the subtree from node y_2

DERIVING A FORWARD EQUATIONS... FOR WHAT?



The analogous quantities at the root,

$g_{e_z}(t_{y_1}) = \mathcal{P}(\mathcal{T}_{y_1} \mid y_1 = z, t_{y_1})$, also do not provide us with information about the transition probability from the root state to the node y_2 .

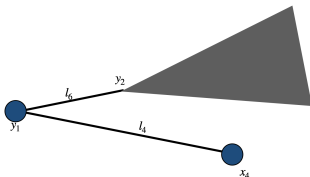
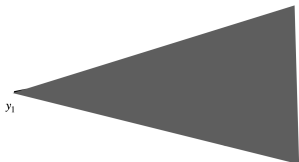
DERIVING A FORWARD EQUATION... FOR WHAT?



We need a different object to describe this **forward** transition

And there's the clue: that something needs to be described by a **forward equation** rather than a backward equation

DEFINE A NEW QUANTITY

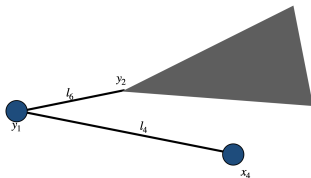
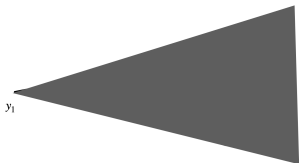


Let's consider the probability of that an edge e is in state i at time t , given the state of its ancestral node at some earlier time:

$$\tilde{g}_{e_i}(t) = \mathcal{P}(\text{lineage } e \text{ alive **and in state } i \text{ at time } t \mid \dots)**$$

Contrast this with what we had before:

$$g_{e_i}(t) = \mathcal{P}(\mathcal{T}_e(s_1) \mid \text{lineage } e \text{ alive **and in state } i \text{ at time } t)**$$

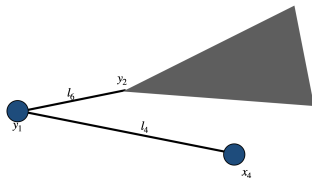
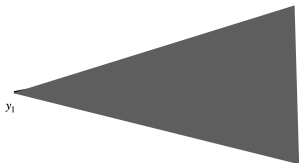
DERIVE A FORWARD EQUATION FOR $\tilde{g}_{e_i}$ 

Then (again, with the annoying convention that $t - \Delta t$ is in the future relative to time t), it seems natural that

$$\mathcal{P}(\text{lineage } e \text{ alive and in state } i \text{ at time } t - \Delta t \mid \dots) = \sum_j \mathcal{P}(\text{lineage } e \text{ alive and in state } j \text{ at time } t \mid \dots) p_{ji}(\Delta t) + \dots$$

Can anything else happen? Edge e needs to stay alive. It also can't produce new lineages with descendants of their own.

DERIVE A FORWARD EQUATION FOR $\tilde{g}_{e_i}$



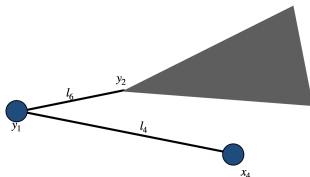
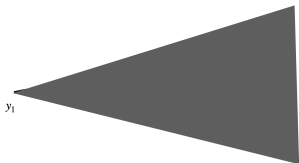
So we could have

$$\begin{aligned}\tilde{g}_{e_i}(t - \Delta t) = & \tilde{g}_{e_i}(t) \left(1 - (\mu_i + \lambda_i + \sum_j q_{ij})\Delta t \right) \\ & + \sum_j \tilde{g}_{e_j}(t) q_{ji} \Delta t + \tilde{g}_{e_i}(t) 2(\lambda \Delta t) E_{e_i}(t),\end{aligned}$$

and E_{e_i} is the backward quantity from before.³

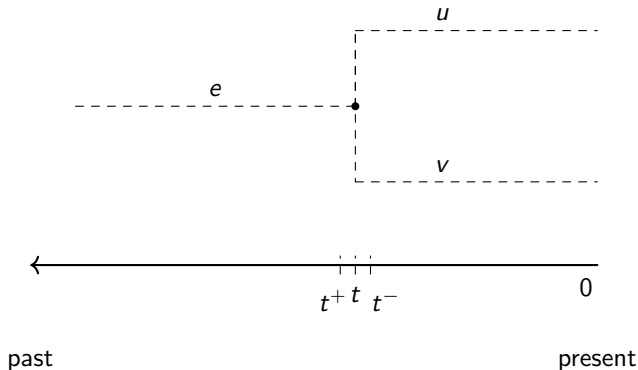
³Yexuan Song et al. "SAASI: Sampling Aware Ancestral State Inference".
In: *Nature Communications* (2026).

TRANSITION PROBABILITY ALONG AN EDGE



We end up with an equation for $\frac{\tilde{g}_{e_i}}{dt}$ that we integrate down the tree, starting from the root and working our way down. We have to be careful about what happens at a branching point.

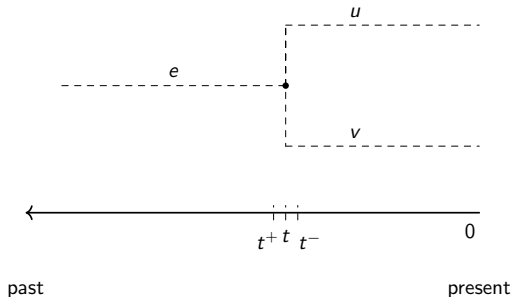
BRANCHING POINTS



From the postorder traversal (initial likelihood calculation), we obtained $g_{u_i}(t^-)$ and $g_{v_i}(t^-)$, which are the conditional likelihoods of the subtrees below those edges.

From the preorder traversal down edge e , we have $\tilde{g}_{e_i}(t^+)$.

BRANCHING POINTS



To proceed, $\tilde{g}_{u_i}$ and $\tilde{g}_{v_i}$ are initialized at t^- according to⁴

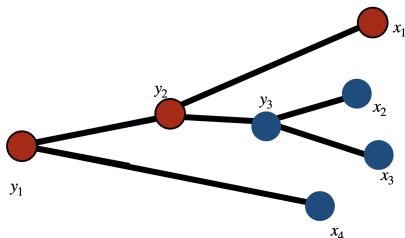
$$\tilde{g}_{u_i}(t^-) = \lambda \tilde{g}_{e_i}(t^+) g_{u_i}(t^-),$$

$$\tilde{g}_{v_i}(t^-) = \lambda \tilde{g}_{e_i}(t^+) g_{v_i}(t^-).$$

Question: why not initialize $\tilde{g}_{u_i}$ by applying argmax_i to $\lambda \tilde{g}_{e_i}(t^+) g_{u_i}(t^-)$, as in Pupko et al. (2000)?

⁴Yexuan Song et al. "SAASI: Sampling Aware Ancestral State Inference".
In: *Nature Communications* (2026).

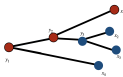
TRANSITION PROBABILITY ALONG AN EDGE



Once the preorder traversal is finished, $\frac{\tilde{g}_{e_i}(t)g_{e_i}(t)}{\sum_i \tilde{g}_{e_i}(t)g_{e_i}(t)}$ is calculated at each internal node.

A joint reconstruction is obtained by evaluating argmax wrt i .

SUMMARY



- We saw Stadler's approach for tree probabilities/likelihoods in the sampling-through-time model extends to SSE models
- This uses backward equations in a postorder traversal
- The tree likelihood computations are still the dynamic programming/Felsenstein pruning algorithm, just harder to calculate
- To do joint ancestral reconstruction, we modify the algorithm from Pupko et al. (2000). We need forward equations to do the preorder traversal
- **Question** Is it a Viterbi algorithm, or a forward/backward algorithm?